

K872 — RATIFY two Grace findings: (1) the α -ladder unification — the inter-stratum overlap between adjacent Bergman levels = α (Shilov integral, k-independent; BST_ElectronMass_CanonicalProof Step 3), so Lane 1 (mass ratios) and Lane 3 (scale) are ONE machine (verified). (2) the coordinate-inconsistency catch — the muon $e \rightarrow \mu$ distance is 0.542 α -steps (2.666 nats), NON-integer (naive α -power gives $\alpha^{-2}=18,779$ vs 207, 91 $\times$ off), AND the electron is placed inconsistently ($\nu=5/2$ continuum in the Wallach picture vs Bergman weight $k=1$ below-threshold in the mass proof). Likely resolution: k (Bergman/K-type weight) and ν (Wallach/position) are DIFFERENT variables; the generations live in ν ; the $k \leftrightarrow \nu$ dictionary is UNPINNED — so the failed ratio attempts were differencing incompatible coordinates. The unblock is a DEFINITIONAL fix (source the $k \leftrightarrow \nu$ dictionary), NOT a 9th reframe — it makes the computation well-posed. The structural bank is coordinate-INDEPENDENT (a phase count) and SURVIVES. The values have collapsed to ONE open number: the two-point Bergman distance in the right coordinate.

Keeper | 2026-07-24 Fri | Ratify the unification + the coordinate catch; classify the unblock as definitional not a reframe (stopping-rule relevant). Plain.

(1) The α -ladder unification — RATIFIED, banked (Grace)

- The overlap between adjacent Bergman levels = α (the fine-structure constant), from the Shilov-boundary integral, k-independent (BST_ElectronMass_CanonicalProof, Step 3).
- $\square$ the muon seesaw (ratios, Lane 1) and the α -tower (scale, Lane 3) are the SAME machine. Two of the three value-lanes collapse into one. This is a real structural unification and it stands regardless of whether the muon value derives.
- Updated lane count: ratios+scale = one α -ladder (Lanes 1+3); mixing/neutrino separate (Lane 2); structure banked. Cleaner than K870/K871's three lanes.

(2) The coordinate-inconsistency catch — RATIFIED (the likely reason for 2 days of non-convergence)

- The α -ladder is non-integer: $e \rightarrow \mu = 0.5 \cdot \ln(m_\mu/m_e) = 2.666$ nats = 0.542 α -steps (one α -step = $\ln(1/\alpha) = 4.920$ nats). The naive integer reading ($\alpha^{-2} = 18,779$) is 91 $\times$ off 207. So the muon ratio is NOT a clean α -power. Verified.
- The electron is placed in two incompatible coordinates: $\nu=5/2$ (ρ_1 , CONTINUUM, above threshold $a/2=3/2$) in the Wallach picture; Bergman weight $k=1$ (BELOW the square-integrable threshold) in the electron-mass proof. Above vs below the threshold — contradictory IF k and ν are the same variable.
- Grace's deeper catch — the positions $\{5/2, 3/2, 0\}$ mix THREE coordinate systems: ρ -components (5/2, 3/2), Wallach points $\{0, 3/2\}$ (share only 3/2), and Bergman weights k (electron at $k=1$). No dictionary pins them. The failed ratio attempts (including this morning's) were differencing incompatible coordinates — very likely why the number kept coming out wrong.

★ The audit resolution — a DEFINITIONAL fix, not a reframe

- Likely: k (Bergman/K-type weight) and ν (Wallach/position parameter) are different variables. The generations live in the position $\nu = \{5/2, 3/2, 0\}$ (Wallach strata); the Bergman weight $k=1$ for the electron is a separate index; the $k \leftrightarrow \nu$ dictionary is unpinned. Mass may depend on both, so differencing across them is ill-defined until the dictionary is sourced.
- This is a definitional/consistency fix (rep-theory sourcing), NOT a 9th mass-formula attempt. It makes the computation WELL-POSED. Distinguish sharply from the 8 reframes (which were new formulas) — this is fixing the coordinate the formulas were (mis)using. So it does NOT trip the stopping rule; it's the prerequisite the stopping-rule test (the two-point distance) needs.
- Corpus has partial K-type structure (Lyra F338 E0-convention flag, L4 mass anchor) but the $k \leftrightarrow \nu$ dictionary is not pinned — needs the book (weighted-Bergman / Faraut-Koranyi), not memory.

What SURVIVES (coordinate-independent) — the structural bank

The classification — the 3 generations occupy the 3 Wallach PHASES (continuum / discrete-3/2 / discrete-0) — is a PHASE COUNT, not a coordinate difference. why-exactly-3 = rank+1, no-4th, Principle #16. Coordinate-independent → SAFE, banked. The paper (F676 + Grace's render) is untouched.

The values, honestly localized (this is GOOD)

“Values resist” has collapsed to a SINGLE open number: - Scale (Lane 3): nearly derived ($6\pi^5$ F402 + $2C_2$ F426 + per-step- α , K871). - Shared per-step coupling = $\alpha = N_{\max} + 1/(2gr)$: BST-sourced, not open (Elie closed my K871 ask). - Unification (Lanes 1+3 = α -ladder): banked (Grace). - Ratio (Lane 1): the ONE open number = the two-point Bergman distance $\mu \leftrightarrow e$ (2.666 nats) in the RIGHT coordinate. Blocked on (a) the $k \leftrightarrow \nu$ dictionary [definitional], then (b) the two-point weighted-Bergman kernel distance [FK book]. Naive reading fails 91×; leans structural (F585 Tier-2 floor) unless the sourced distance gives 2.666 with no free scale.

Stopping rule (K870) — where we are

The naive α -ladder reading FAILED (91×). Per the rule, that would tier the ratio structural — BUT Grace's coordinate catch shows the failure was an ill-posed coordinate, not a genuine negative. So: the coordinate fix (definitional) is a legitimate unblock; after it, the two-point Bergman distance is THE last clean test. If it sources to 2.666 with no chosen scale → the ratio derives. If it needs a chosen scale → tier structural (identified). No new formulas — this is the same off-diagonal α -ladder object, computed in the right coordinate.

Directions

- ★ GRACE (lead): the $k \leftrightarrow \nu$ dictionary — source (weighted-Bergman/FK book) which coordinate the generations live in and the $k \leftrightarrow \nu$ map. This is the definitional unblock; it makes the two-point distance well-posed. Elie's offered “target-innocent discriminator for $k=1$ -vs- $5/2$ electron assignment” feeds this directly — take it.
- ★ ELIE: build the $k=1$ -vs- $5/2$ discriminator (your offer) — it's exactly the coordinate-fix input. Then W4/cross-check fires on the sourced two-point distance.
- ★ LYRA: fold Grace's render into F676 (durable, ready) + Lane 2 (neutrino $m_1=0$, $|U_{e2}|^2$ — the same overlap machinery, per your synthesis). The mixing and the ratio are one machine once the coordinate is pinned.
- KEEPER: α -ladder unification + coordinate catch RATIFIED; the unblock is definitional (not a reframe, stopping-rule-safe); structure coordinate-independent/safe; values localized to one number (the two-point distance in the right coordinate).

— Keeper K872, 2026-07-24. RATIFIED: (1) α -ladder unification — inter-stratum overlap = $\alpha \rightarrow$ Lanes 1(ratios)+3(scale) = ONE machine (Grace, BST_ElectronMass_CanonicalProof). (2) coordinate-inconsistency catch — muon $e \rightarrow \mu = 0.542$ α -steps (2.666 nats, non-integer; naive $\alpha^{-2}=18,779$ vs 207, 91×); electron placed inconsistently ($\nu=5/2$ continuum vs Bergman $k=1$ below-threshold); positions $\{5/2, 3/2, 0\}$ mix 3 coordinate systems

(ρ -components/Wallach/Bergman-k), dictionary unpinned $\rightarrow$ failed ratios differenced incompatible coordinates. Resolution: k and ν are DIFFERENT variables, generations live in ν , $k \leftrightarrow \nu$ dictionary needs sourcing — a DEFINITIONAL unblock, NOT a 9th reframe (stopping-rule-safe). Structure = phase count, coordinate-independent, SAFE. Values localized to ONE number: the two-point Bergman distance $\mu \leftrightarrow e$ in the right coordinate. See [[Keeper_K871_CORRECTION_to_K870_Lane3_scale_is_NEARLY_DERIVED_not_the_bottleneck_F402_prefactor_F426_expo 07-24]], [[Keeper_K870_META_two_day_pattern_structure_banks_values_resist_the_8_reframes_split_diagonal_vs_offdiagonal 07-24]], Grace (α -ladder + coordinate catch), BST_ElectronMass_CanonicalProof (overlap= α), Lyra F338/L4 (K-type structure).